

5.6 χ -bounded graph properties

Our motivation for considering perfect graphs in the last section was that high chromaticity is easier to understand for these graphs than in general: if a perfect graph has chromatic number at least r , it has a local reason for this in the form of a K^r subgraph. It turned out that perfection has structural implications far beyond this. We may ask, therefore, whether graph properties defined by a slightly weaker condition might still imply some interesting structure for the graphs that have them.

χ -bounded
 χ -bounding

Consider any graph property $\mathcal{G}$ which, like perfection, is closed under taking induced subgraphs: a class that contains all induced subgraphs of its elements. We call $\mathcal{G}$ χ -bounded if there exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$, called a χ -bounding function, such that $\chi(G) \leq f(r)$ for all graphs $G \in \mathcal{G}$ of clique number $\omega(G) \leq r$, for all $r \in \mathbb{N}$. In particular, then, $\chi(G) \leq f(\omega(G))$; but note that for individual functions f this is a weaker condition. Perfection, for example, is χ -bounded with χ -bounding function $f(r) = r$.

Qualitatively, graphs in a χ -bounded class $\mathcal{G}$ have complete subgraphs as large as desired as soon as their chromatic number is big enough: that is the idea behind the notion of χ -boundedness. Quantitatively, if f is a χ -bounding function for $\mathcal{G}$, then any $G \in \mathcal{G}$ with $\chi(G) > f(r)$ contains a K^{r+1} , for every $r \in \mathbb{N}$.

Not all graph classes that are closed under taking induced subgraphs are χ -bounded: the triangle-free graphs, for example, are not, by Erdős's Theorem 5.2.5. So which classes are?

Every graph property $\mathcal{G}$ that is closed under induced subgraphs is characterized by the class $\mathcal{H}$ of graphs that are not in $\mathcal{G}$ and are minimal with this property under the induced-subgraph relation: an arbitrary graph G lies in $\mathcal{G}$ if and only if it has no induced subgraph in $\mathcal{H}$. One way to search for χ -bounded graph properties, therefore, is to examine those characterized in this way by some particularly simple classes $\mathcal{H}$.

The strong perfect graph theorem implies that $\mathcal{G}$ is χ -bounded when $\mathcal{H}$ consists of all odd holes and antiholes. But since χ -boundedness is weaker than perfection, we can make $\mathcal{H}$ smaller, thereby making $\mathcal{G}$ larger:

Theorem 5.6.1. (Scott & Seymour 2016)
The graphs with no odd hole are χ -bounded.

Theorem 5.6.1 is one of the earliest results of its kind proved in the wake of the strong perfect graph theorem. It has since been strengthened in various ways, see the notes. The notion of χ -boundedness itself is much older. It was introduced by Gyárfás in 1975, who later proved the following pretty result:

Proposition 5.6.2. *If P is any fixed path, then the graphs not containing P as an induced subgraph are χ -bounded.*

We shall derive Proposition 5.6.2 from a lemma that weakens the assumption of not containing the given path to better support induction:

Lemma 5.6.3. *There is a function $g: \mathbb{N}^2 \rightarrow \mathbb{N}$ such that $\chi(G) \leq g(c, k)$ for every connected graph G satisfying*

- (i) $\chi(G[N(v)]) \leq c$ for every vertex $v \in G$;
- (ii) G has a vertex v at which no induced path of length k starts.

Proof. We prove by induction on k that the function g defined recursively by $g(c, 1) := 1$ and $g(c, k) := g(c, k-1) + c + 1$, for all $c \in \mathbb{N}$ and $k > 1$, is as desired. The induction starts with $k = 1$, in which case $G = K^1$ by (ii).

For the induction step pick $v \in G$ as in (ii). Since G is connected, every component C of $G - N(v) - v$ has a neighbour u in $N(v)$. By induction, $\chi(G[V(C) \cup \{u\}]) \leq g(c, k-1)$, so also $\chi(C) \leq g(c, k-1)$. Combining such colourings of all these C with a $(c+1)$ -colouring of v and its neighbours, which exists by (i), yields a $g(c, k)$ -colouring of G . $\square$

Proof of Proposition 5.6.2. Fix any $k \in \mathbb{N}$. Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be defined recursively by $f(1) := 1$ and $f(r) := g(f(r-1), k)$ for $r > 1$, where g is the function from Lemma 5.6.3. We prove by induction on r that every graph G that contains no path of length k induced and is such that $\omega(G) \leq r$ satisfies $\chi(G) \leq f(r)$.

The induction starts at $r = 1$, in which case G consists of isolated vertices. For the induction step let $G_v := G[N(v)]$ for vertices v of G . As $\omega(G) \leq r$ we have $\omega(G_v) \leq r-1$, and hence $\chi(G_v) \leq f(r-1)$, for every v . By Lemma 5.6.3 with $c = f(r-1)$ this yields $\chi(G) \leq f(r)$. $\square$

Proposition 5.6.2 is a special case of a more sweeping conjecture. As we noted earlier for triangles, excluding a graph H will not define a χ -bounded graph property if H contains a cycle: by Theorem 5.2.5 there are graphs of arbitrarily large chromatic number and girth greater than $g(H)$, which thus do not contain H but have clique number 2.

The following conjecture claims that this is the only obstruction to χ -boundedness: that excluding any graph that does *not* contain a cycle will define a χ -bounded class.

Conjecture. (Gyárfás 1975; Sumner 1981)

If F is any fixed forest, then the graphs not containing F as an induced subgraph are χ -bounded.

Exercises

- 1.⁻ Show that the four colour theorem does indeed solve the map colouring problem stated in the first sentence of the chapter. Conversely, does the 4-colourability of every map imply the four colour theorem?
 - 2.⁻ Show that, for the map colouring problem above, it suffices to consider maps such that no point lies on the boundary of more than three countries. How does this affect the proof of the four colour theorem?
 3. Try to turn the proof of the five colour theorem into one of the four colour theorem, as follows. Defining v and H as before, assume inductively that H has a 4-colouring; then proceed as before. Where does the proof fail?
 4. Calculate the chromatic number of a graph in terms of the chromatic numbers of its blocks.
 - 5.⁻ Let G be a graph, and let $k \in \mathbb{N}$.
 - (i) Show that G has chromatic number at most k if and only if there exists a homomorphism from G to K^k .
 - (ii) Show that G is bipartite if and only if there exists a homomorphism from G to K^2 or to an even cycle.
 - (iii) Are there homomorphisms from C^{17} to C^7 , from C^7 to C^{17} , from C^{16} to C^7 , and from C^{17} to C^6 ?
 6. Show that graphs of large girth and without a given minor are ‘nearly bipartite’ in the following sense. Let H be a fixed graph and C a fixed odd cycle. Use Theorem 7.2.6 to show that if G is a graph of sufficiently large girth (depending only on H and C) that does not contain H as a minor, then there is a homomorphism from G to C .
 7. For every $n > 1$, find a bipartite graph on $2n$ vertices, ordered in such a way that the greedy algorithm uses n rather than 2 colours.
 8. Consider the following approach to vertex colouring. First, find a maximal independent set of vertices and colour these with colour 1; then find a maximal independent set of vertices in the remaining graph and colour those 2, and so on. Compare this algorithm with the greedy algorithm: which is better?
 9. Show that the bound of Proposition 5.2.2 is always at least as sharp as that of Proposition 5.2.1.
- A k -chromatic graph G is called *critically k -chromatic*, or just *critical*, if $\chi(G - v) < k$ for every $v \in V(G)$.
10. Determine the critical 3-chromatic graphs.
 - 11.⁺ Show that every critical k -chromatic graph is $(k - 1)$ -edge-connected.
 12. Formalize and prove the following statement: assuming large average degree drives the colouring number up but not the chromatic number.